

PRASANNA KUMAR MANDAPATI

Senior Java Full Stack Engineer | Backend Systems | Distributed Services | AI/RAG Workflows

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PROFESSIONAL SUMMARY

Java Full Stack Engineer with 5+ years of experience building production-grade backend systems, distributed services, APIs, batch pipelines, and enterprise integrations across automotive, vendor operations, and middleware platforms. Strong backend depth in Java 17/21, Spring Boot, WebFlux, Spring Batch, Kafka, PostgreSQL, MongoDB, AWS, Kubernetes, and CI/CD, with React/Angular frontend integration experience. Current GM work includes GFEP 2.1 manufacturing execution workflows and eDiagnostics, deployed for North America, supporting vehicle diagnostic program setup, production readiness, validation, and 2D/3D visualization data. Experienced with Python FastAPI RAG workflows, vector search, embeddings, and LLM-assisted diagnostic knowledge retrieval. Known for clean service boundaries, scalable design, reliable delivery, and hands-on production support.

TECHNICAL SKILLS

Languages: Java 17/21, Python, JavaScript, TypeScript, SQL, Scala, ESQL

Backend: Spring Boot, Spring WebFlux, Spring MVC, Spring Batch, Spring Security, JPA/Hibernate, REST APIs, Microservices, Maven

Frontend: React.js, Angular, TypeScript, JavaScript, HTML5, CSS3, API-driven dashboards and validation screens

Messaging/Data: Kafka, RabbitMQ, IBM MQ, AWS SQS/SNS, PostgreSQL, MongoDB, Oracle, MySQL, SQL Server, Redis

Cloud/DevOps: AWS EC2/S3, GCP, Docker, Kubernetes, OpenShift, Artifactory Cloud, Spin Cloud, Jenkins, GitHub Actions, Argo

AI/RAG: Python FastAPI, RAG pipelines, embeddings, vector search, query transformation, prompt engineering, LLM response validation

Quality/Ops: JUnit, Mockito, cucumber, Postman, SonarQube, CodeQL, Splunk, Grafana, Dynatrace, JVM metrics, Kafka lag checks

PROFESSIONAL EXPERIENCE

Algebra IT - General Motors, Austin, TX | Senior Software Engineer

Mar 2025 - Present

Project 1: GFEP 2.1 - Global Factory Execution Platform | Current Project Enterprise manufacturing execution platform supporting factory workflows, program execution, operational tracking, and production data processing.

- Design and develop backend microservices using Java 17/21, Spring Boot, Spring WebFlux, and REST APIs for high-volume manufacturing execution and program data workflows.
- Build APIs for program setup, production tracking, operational status lookup, validation results, and downstream integration workflows used by internal teams.
- Implement Kafka-based asynchronous processing to decouple long-running manufacturing workflows from user-facing APIs and improve service resiliency.
- Create idempotent Kafka consumers with retry handling and dead-letter patterns to safely process replayed, delayed, or failed program events.
- Develop Spring Batch jobs for ingestion, validation, transformation, and persistence of large JSON and enterprise data files into PostgreSQL.
- Improve backend performance through optimized SQL queries, batch inserts, indexing strategies, connection handling, and targeted JVM tuning.
- Support Kubernetes deployments with readiness probes, health checks, config maps, secrets, resource limits, rollout validation, and production monitoring in Splunk.

Project 2: eDiagnostics Web-Based Application | Completed and Deployed for North America Web platform used by service dealers and internal teams to manage vehicle diagnostic programs, production readiness, and 2D/3D visualization data.

- Built Java/Spring Boot, WebFlux, Spring Batch, Kafka, and PostgreSQL services for diagnostic program setup, validation, and production-readiness workflows.
- Developed REST APIs for vehicle search, component layout retrieval, circuit diagram lookup, VIN schematic views, and wiring harness inspection.
- Designed ingestion pipelines to process large Teamcenter and enterprise JSON/Excel datasets into PostgreSQL with audit fields such as created_by and created_date.
- Created validation logic and reports to detect missing engineering codes, inconsistent service descriptions, unsupported mappings, and wiring-harness data gaps.
- Improved diagnostic data reliability from roughly 60% to over 95% by strengthening validation rules, re-ingestion workflows, and data-quality checks.
- Added parallel processing, retry/fallback logic, and alternate-source recovery to reduce ingestion time and improve stability when upstream systems were unavailable.
- Integrated AWS S3 and Artifactory Cloud for processed diagnostic files, JSON outputs, application artifacts, and downstream program data.
- Supported North America deployment, post-release validation, production triage, and cross-team coordination with QA, frontend, DevOps, and engineering users.

AI/RAG Workflows at GM: Integrated Java backend services with a Python FastAPI RAG service for diagnostic knowledge retrieval, using document chunking, embeddings, metadata filtering, query transformation, reranking, and source-backed LLM responses. Added guardrails so AI answers stayed grounded in approved runbooks, wiring references, known issue notes, and engineering mapping rules.

Algebra IT - Apple Inc., Austin, TX | Software Engineer

Jun 2023 - Feb 2025

Project: Vendor Vue Web-Based Application Vendor operations platform for employee hours, vendor transactions, job processing, dependencies, and payment workflows.

- Designed and developed Java 21, Spring Boot, Spring WebFlux, and REST API microservices for vendor, employee, payment, and job-processing workflows.
- Created APIs supporting React.js screens for employee hour tracking, vendor transaction review, payment status lookup, and operational reporting.
- Implemented Spring Batch ETL pipelines for datasets exceeding 1M records across validation, transformation, enrichment, and MongoDB persistence workflows.
- Built Kafka-based asynchronous workflows for vendor transaction updates, payment status changes, batch job notifications, and downstream dependency coordination.
- Developed Kafka consumers with retry handling, offset control, and dead-letter routing to keep asynchronous processing reliable during downstream delays.
- Built MongoDB aggregation pipelines for complex reporting, filtering, analytics, and operational views across vendor and employee data.
- Improved system performance by 20% through backend tuning, query optimization, reduced service calls, and improved transaction-processing paths.
- Deployed services using AWS EC2, AWS S3, and Spin Cloud, supported 99.9% uptime, and used Splunk to debug APIs, batch jobs, Kafka gaps, and production issues.

Agility E Services Pvt Ltd, India | Software Developer

Jun 2019 - Jul 2021

Project: Enterprise Integration and e-Invoicing Platform Enterprise integration platform for financial and retail clients using Java, Spring Boot, IBM MQ/IIB, REST APIs, secure file exchange, and compliance workflows.

- Engineered Java, Spring Boot, IBM MQ, and IBM Integration Bus flows to automate data movement between legacy systems and external applications.
- Built REST-based backend integration services to replace point-to-point flows, improving data exchange speed and system stability.
- Designed secure SFTP/FTP modules and middleware transformation workflows for reliable business-critical file movement across platforms.
- Developed Java services for e-invoice generation, IRP portal integration, QR-code generation, and compliance-oriented data transformation.

- Improved operational efficiency by 25-30% through automation, structured logging, stronger exception handling, GitHub migration, and better production monitoring.

ENGINEERING STRENGTHS

- Backend systems: clean controller-service-repository boundaries, reusable validation logic, stable REST contracts, consistent exception mapping, and maintainable code.
- Distributed processing: Kafka, Spring Batch, and queue-based workflows to decouple long-running work, support safe retries, avoid API blocking, and improve resiliency.
- Performance and production: SQL tuning, indexing, batch inserts, JVM metrics, Kafka lag checks, Splunk analysis, API latency debugging, and root-cause investigation.
- Quality and delivery: JUnit, Mockito, Postman, SonarQube, CodeQL, pull request reviews, CI/CD validation, Agile delivery, and mentoring junior developers.
- Frontend integration: React.js/Angular API-driven dashboards, validation flows, reporting screens, operational views, and clear error-state handling.

PROJECT AND SYSTEMS HIGHLIGHTS

- AI-assisted diagnostic RAG workflow: Java backend integrates with Python FastAPI retrieval service to search runbooks, wiring references, API docs, and known issue notes using embeddings, metadata filters, and source-backed LLM responses.
- Vehicle data validation platform: batch and API workflows validate Teamcenter/enterprise diagnostic datasets, generate actionable error reports, and support clean re-ingestion for production readiness.
- Vendor operations event platform: Spring Batch, Kafka, MongoDB, and React workflows process high-volume vendor transactions, employee-hour data, payment status events, and operational reports.
- Enterprise integration platform: Java, IBM MQ/IIB, REST APIs, and secure file exchange automate data movement, compliance transformations, e-invoice generation, and legacy modernization.
- CS fundamentals used in daily work: object-oriented design, data structures, algorithmic problem solving, concurrency concepts, database systems, distributed systems, API design, and reliability engineering.

TECHNICAL LEADERSHIP AND DELIVERY

- Own features end-to-end: clarify requirements, design APIs, implement backend logic, support frontend integration, test changes, deploy safely, and validate production behavior.
- Lead by delivery: review pull requests, mentor developers, document runbooks, communicate blockers early, and help teams split large work into reliable incremental releases.
- Build for maintainability: prefer simple service boundaries, explicit data contracts, clear logs, repeatable tests, and operational dashboards over fragile one-off fixes.
- Work well in ambiguity: read unfamiliar enterprise codebases, isolate production defects, coordinate across platform teams, and ship safe changes without risky rewrites.

EDUCATION

Master of Science in Data Science Indiana University Purdue University	May 2023
Bachelor of Technology in Computer Science and Engineering Bapatla Engineering College	Apr 2019

CERTIFICATION

[AWS Certified Developer Associate](#)